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Model for Cutting Tools Usage Tracking by On-line Data Capturing and Analysis

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Abstract

A model to track the cutting usage has been developed. The application domain of the model is milling in the woodwork industry. The model involves the capturing of raw data from three different sources; the machine tool control, the Enterprise-Resource-Planning system and the tool management system. The data is analysed, the cutting distance is calculated and stored along with the data of the tool usage parameter settings. In order to predict the remaining tool life it is essential to know the actual tool usage. The usage is based on the tool life distance which is calculated by the movement data of the axis and various parameters of the numerical control system. Input for the calculation is the cutting path, the tool usage data and the end of life due to tool wear. With this information it is possible to forecast the estimated tool life by using machine learning algorithms. This forecast algorithm will predict more accurate results after each learning cycle. The developed model has been implemented as a smart service. The tool wear predictions are used for improvement of machine availability. Tool changes can be done in advance of large orders and machining can be operated with less supervision. Furthermore the shipping of tools between the manufacturers, the customer and the refurbishment can be optimised. Premature wear due to wrong settings or human error can be avoided. The system ensures a reliable exchange of information without paper work via the internet across company borders for the whole life cycle of each tool.

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1. Introduction and Motivation

Tool wear is not preventable in chip-forming machining. The chain of causation between cutting conditions and tool life has been empirically evaluated for certain process parameters. At these certain parameters the prediction of the remaining tool life is accurate. With different process parameters a calculated and estimated prediction is less accurate.

The correlation between additional process parameters could be determined with a high number of lab test runs. An alternative approach is to capture and analyse the tool operation data and the resulting wear.

New developments in manufacturing are machine integration and automated transmission of data [1]. The automated transmission of tool machine control data allows reasoning of the tool operation conditions. This data can be set into context of other production data such as process planning

data and tool management. The combination of these different data sources create higher information out of the raw data.

Surveys have shown that tool costs are representing up to 8% of the total manufacturing costs. Due to personnel costs or machine down times because of wrong or missing tools, the number can increase up to 34% [2]. Therefore it is essential for the process planner to know the tool life of tool under certain cutting circumstances and plan tool replacement before tool wear results in bad quality parts or tool breakage.

Research of tool life has been conducted since the beginning of the 20th century. In 1907 Taylor [3] proved there is a correlation between cutting speed and tool life. This correlation and various other influencing factors on tool life have been studied extensively. All these studies were conducted in lab environments. The lab environment and conditions differ from the usage on the shop floor. The challenge is an application and adaption of these models in a real production environment

where the individual wear of thousands of tools has to be predicted.

Therefore new approaches and models have to be developed to overcome these challenge.

2. State of the Art

State of the art in tool management, wear models and tool life prediction are considered.

2.1. State of the Art in Tool Management

Tool management systems are required to plan, control and monitor the tool material and information flow to increase the transparency in production systems and reduce tool costs.

Without tool management systems, machine operators need almost 20% of their working time to search tools and 15% of the manufacturing orders are late due to missing tools [4]. These systems can be divided into 5 main functions:

- tool planning
- tool management
- tool scheduling
- tool supply
- tool usage

The tool planning consist mainly of order-independent functions to plan the tool range for future processing tasks. The tool management works on tool procurement, tool inventory management and tool requirement planning. To provide the right tools at the right time in the correct number and quality for the planned processing, tool scheduling has to be planned.

Due to the fact some tools get reconditioned for their next usage, the tool flow can be represented as cycle. The physical tool cycle shows the supply up to the usage on the tool machines. These last two functions of Tool Management System are located on the shop floor [4].

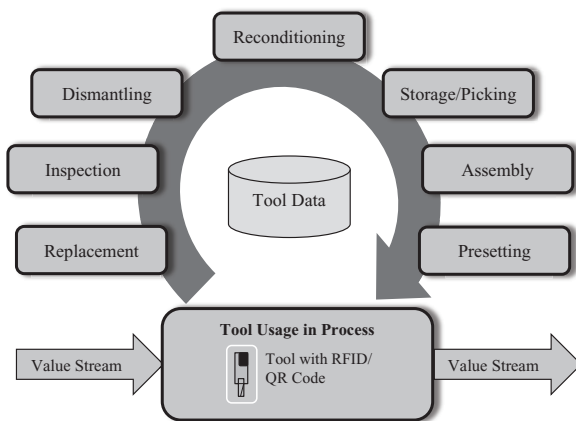


Fig. 1 Physical Tool Cycle

Figure 1 shows the physical tool cycle with all stations. The tool supply is responsible for the tool inspection, the tool dismantling, tool reconditioning, for the tool storage and commissioning, the tool assembly and presetting and finally the tool replacement. To realise an adequate tool flow, transport

systems have to be installed. All actions lead to the main cycle station: the tool usage [5].

Identification systems should support the information flow through the different stations. Various research projects have evaluated and verified the usage of RFID tags or optical codes, such as QR Codes, on the tool holder or on the tool itself [6],[7],[8].

Tool Data Management systems support the Tool Management System by providing actual tool data for all five main functions through the tool life cycle. The information structures can be divided into:

Process data:

- geometrical data
- technological information

Administrative data:

- identification data
- organisational data

Condition and historical data which are partly linked to process and administrative data [9].

2.2. Existing Tool Wear Models

The application domain of this model is the woodwork industry. Therefore the considered models are for wood machining application.

To this date numerous experimental wear studies have been conducted [10],[11]. The generalisation of the conducted experiments was achieved by Fischer [12],[13]. His model explains the cause of the wear and also the wear progression.

By exceeding the specific compression strength or threshold pressure (See Fig 2a) cutting material gets removed.

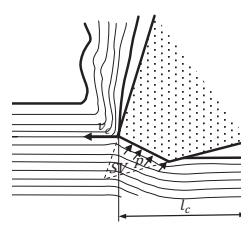


Fig. 2a Wear Model

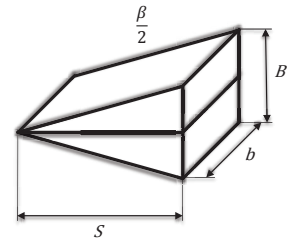


Fig.2b Cutting Edge Variables

The loss of cutting material and the therefore connected loss of volume is constant with constant cutting conditions.

2.3. Tool Life Prediction

Tool life testing can be performed with specific cutting parameters and a defined setup [14]. The tool life is measured with this process parameter setting. The tool life with process parameter variations within the process field can only be determined with numerous similar test runs. The cutting conditions are constantly different for milling with changes in the tool path direction. State of the art in the industry is the counting of the parts which can be manufactured before the cutting tool reaches the end of life criteria.

This practice is not useable with high part variations.

Alternative approaches are the measuring of the spindle power or torque monitoring. This can be done with the help of force sensors. With the force information the tool end of life criteria can be predicted [15].

3. Approach

Nowadays new information and communications technology enable the collection of large datasets during the in-process usage of tools. The data of tool machines motion and parameter settings in the machine control was unused for tool wear predictions, but now can be transmitted via public global networks.

The tool usage for production purpose differs compared to lab test runs for wear. Instead of static values for cutting speed and feed rate the values spread in a range due to the acceleration and deceleration in order to change the tools path. This also leads to a range of feed rate per tooth which has significant influence on wear.

The data transmitted from the tool machine control is more accurate than planned data like the NC-Code due to the fact that as-is parameter settings can differ from the planned settings. This is often the case in feed rate settings which the operator can easily vary in operation.

It is now possible to capture tool operation and combine it with static data and use this information in combination with wear measurements. The wear factors are now possible to be learned.

To implement this approach no additional sensors are required at the tool machine and only software implementations are needed.

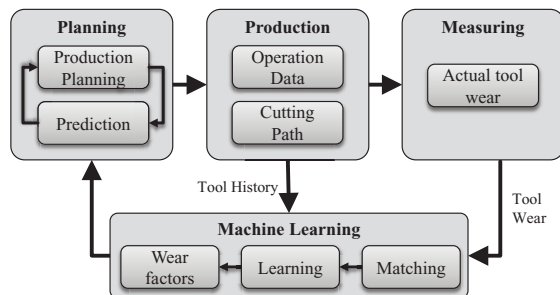


Fig. 3 Concept Overview

The approach of the concept overview is illustrated in Figure 3. The idea consist of capturing tool life history and matching this data with the actual measured wear occurring at the cutter tool. This matched data is the base for the prediction model.

The wear factors from the machine learning are trained at each learning cycle in order to get more accurate predictions.

This paper states a model which incorporates the capturing and analysing of tool usage to calculate tool life prediction based on a machine learning algorithm.

An implementation of this proposed model is also introduced.

The approach begins with an expansion of an existing wear model for woodworking tools. The base is the model developed by Fischer [13]. This model describes the wear in correlation with the cutting path and its cutting conditions. The cutting path is calculated on-the-fly and the history of the tool is recorded. At the end of life the wear is measured and matched with its history. Due to this combination of history and resulting wear machine learning can take place.

3.1. Machine Learning Approach

In this approach the tool cutting path and the individual historical data consisting of the cutting conditions data are gathered and matched with the resulting wear of the tool. The information related to the circumstances are used to update correlation factors based on a machine learning algorithm.

3.2. Wear Model

The Fischer Model [13] explains the correlation between constant cutting conditions with the cutting path and the cutter edge offset (S). Abrasive wear takes place evenly on the exposed surface, See Fig. 2a. The ideal geometry of the cutting edge can be seen in Fig 2b. The volume can be described as seen in Equation 1. The loss of cutting volume explained by Fischer can be set equal to the cutting edge Volume [16].

$$V = c * l_c \quad \text{and} \quad V = 0,5 * B * S * b \quad (1)$$

Now B can be described according to Equation 2.

$$B = 2 * b * S * \tan\left(\frac{\beta}{2}\right) \quad (2)$$

With the elimination of B, S can now be described with constant variables which can be united into k. See Equation 3.

$$S = k * \sqrt{l_c} \quad (3)$$

3.3. Tool Cutting Path

The tool cutting path is the total distance from the entrance into the raw material up until the tool leaves the raw material. The individual interrupted cuts occur on each revolution of the milling cutter. Each cutting edge covers a circular arc path during cutting. This cutting length is calculated along the tool centre path.

3.4. As-Is Data via Machine Integration

The tool machine control needs to be connected via an interface to a calculation service. This approach relies on data refresh rates on a hundredth of a millisecond basis such as graphical user interface (GUI) data.

The As-Is data from the tool machine control is different compared to the planned data. The planned data like NC-Code differs for instance in cutting speed which the operator can change manually anytime.

3.5. Tool Machine Data Storage

The data set has to fulfil two requirements. The first requirement is that the necessary context has to be reproducible. This is essential to match the dataset with the different sources (See Fig.3). The following metadata is stored:

- Tool-ID
- Timestamp
- Machine-ID

The second requirement is to capture the tool history. The tool history reflects its cutting conditions. The variables which accomplish the capturing of the cutting conditions are:

- Material used
- Feed rate per tooth f_z
- Cutting Width a_e
- Cutting Speed v_c
- Cutting Path l_c
- Number of revolutions n

To make this approach work the dataset is recorded with a rate of a hundredth of a millisecond.

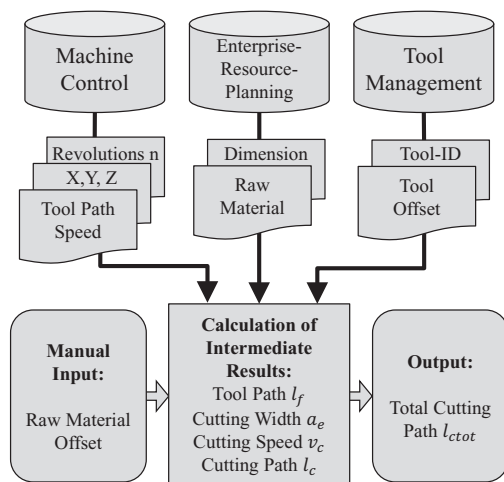


Fig. 4 Data Flow Model

4. Model

For the proposed model the following underlying premises have to be taken into account:

- Tool wear occurs due to abrasion
- No brakeage of cutting tools edges during the operation

Figure 4 shows an overview of the dataflow within the model. The data originates in three different sources and its combination and reasoning is explained in the various steps of the model. Figure 5 states all seven steps of the model.

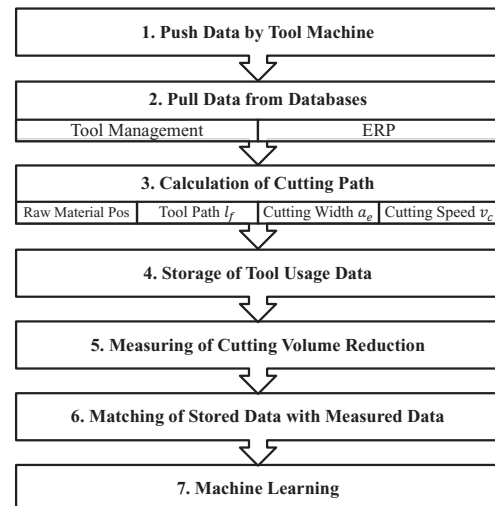


Fig. 5. Steps of the Model

4.1. Push Data by Tool Machine

The model begins with an event-driven data extraction by tool machine control. When there is a change in value from the observed variables after a calculation cycle of the tool machine control, the new value of the variable is passed forward to the calculation and tool history service.

4.2. Pull Data from Databases

The data from the databases is requested by the calculation service. Necessary data from the Tool Management database are Tool-ID and Tool geometry details such as offset variables from its specified size. Data from the ERP system are material type and material dimension specified in the production order.

4.3. Calculation of Cutting Path

The calculation of the cutting path begins with the matching of different coordinate systems. The offset values between the machine zero point and the workpiece zero point have to be known to assess the correct position of the tool in relation to the raw material.

The next step is the calculation of the tool path l_f . This is done with the offset of the XYZ coordinates between the current and the next timestamp. The distance travelled is calculated through the timestamp offset and the cutting speed v_c . The cutting width a_e is calculated by the tool centre point, the tool geometry and the current position of the nearest material.

In case of a smaller cutting width than the tool diameter the circular arc has to be calculated.

Figure 6 shows the circular arc [17]. The nomenclature can be found in the German Standard DIN 6580 [18]. The pressure angle is calculated according to Equation 4

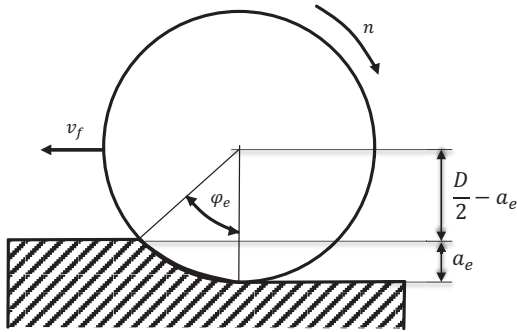


Fig. 6. Pressure Angle φ_e

$$\varphi_e = \arccos \left(\frac{1-2 \cdot a_e}{D} \right) \quad (4)$$

With the pressure angle the circular arc can be calculated according to Equation 5.

$$b = r \cdot \alpha = \frac{D}{2} \cdot \varphi_e \quad (5)$$

There are a number of cuts per tool path segment. The amount of cuts is shown in Equation 6

$$n_c = \frac{l_f}{f_z} = \frac{l_f \cdot n \cdot z}{v_f} \quad (6)$$

The overall cutting path per tool path segment can be calculated according to Equation 7.

$$l_{c \text{ tot}} = \frac{D \cdot \arccos \left(\frac{1-2 \cdot a_e}{D} \right) \cdot l_f \cdot n \cdot z}{2 \cdot 1000 \cdot v_f} \quad (7)$$

Once the calculation is complete the reduced material dimension can be calculated and stored for the next calculation cycle.

4.4. Storage of Tool Usage Data

The data is stored for each tool path segment with timestamp as its primary key. The history of the tool usage can now be accessed for each operation.

4.5. Measuring of Cutting Volume Reduction

The tool edge offset (S) is measured by comparing the original cutting edge dimension with the new cutting edge position due to the volume reduction.

4.6. Matching of Stored Data with Measured Data

The left-hand side of Equation 8 is the measured tool wear and the right-hand side is the cutting condition as part of k and

the calculated cutting path. Both sides of Equation 8 are therefore known.

$$S \sim k \cdot \sqrt{l_c} \quad (8)$$

The separation of the tool history in different tool usage operations is necessary, for instance due to changing raw materials processed.

4.7. Machine Learning and Prediction

The cutting condition data can be broken down into Material M , Feed Rate per Tooth f_z , Cutting Width a_e , Number of Revolutions N , Cutting Speed v_c . According to the Fischer Model [12] the wear volume (S) correlates with the square root of the cutting path. The correlation factor is constant in case of constant cutting circumstances. With variations in the cutting conditions the correlation factor k is subject to a different correlation. This correlation is a function of the input variable. For each input variable a wear factor is introduced in the model, see Equation 9.

$$S \sim k \cdot \sqrt{l_c} \quad \text{with} \quad k = f(k_M, k_{f_z}, k_{a_e}, k_n, k_{v_c}) \quad (9)$$

With the matched data from the wear measurements and the tool history the function of these wear factors can be learned over time. The starting conditions of the function are known for some interactions such as variation of cutting speed or cutting width but no other parameter changes [10],[12],[13].

The wear factors are then used to calculate the remaining tool life. The predicted tool life also differs with the material and the cutting circumstances of the production orders which are going to be processed.

5. Implementation Concept

A prototype implementation was performed. During the implementation the main challenge were the interface to integrate the machine tool into an IP-based network connection with a meta information data model to capture the necessary parameters which otherwise are lost after each calculation cycle of the tool machine control. This is due to the closed proprietary protocols of the tool machine vendors.

Another requirement is the matching of the data stream from the tool machine with the current tool in operation and with the current production order. The matching of the production order is relevant for instance due to the different raw material used.

Another crucial demand is the biunique identification of the tool used in the current operation. Tools have to be distinguishable between sister tools of the same type. It has to be the unique tool to record its own operation history.

The machine integration was achieved by MTConnect [19]. The data is transmitted via Webservice to a Service hosted on a Wildfly Application Server. The Tool history data is stored in a „MongoDB“ and the Tool information administrated by a proprietary Tool Management Software Suite by Leitz GmbH & Co. KG. The graphical user interface is within a web application.

6. Conclusion

Tool operation circumstances are mainly responsible for tool wear. The tool operation conditions can be captured and analysed in regard to its cutting path. To capture this data machine integration is essential. The cutting path has a correlation with the tool wear. This correlation can be learned if tool wear is measured and matched with its history.

With the help of machine learning accurate wear factors are achieved. These wear factors allow insight in the cause and effect chain of cutting conditions and tool wear within the whole parameter set used in production environment.

With the prediction model premature wear due to wrong settings or human error can be avoided. Optimal cutting conditions can extend tool life substantially.

Also planning tasks in tool management can be optimised. For instance stocks levels can be lowered due to the precise tool life prediction.

The model shown needs no additional sensors and can be implemented on a software basis to the tool machine control.

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